

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 2– Concept Mapping (5QUESTIONS)

Course Code:	ITA05	Course Title:	Computer Vision
CO Assessed:	CO1 – Imitation (BL3)	Assessment:	Assessment Tool 2 – Concept Mapping
Weightage:	30% of CIA	Total Marks:	50 (5Questions × 10Marks)
CO1 Statement:	Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems.		
Student Name:	Charali Chandu	Reg. No.:	192472332
Section / Batch:	Slot D / 2024	Date:	14-07-26

2

Code of Conduct

I Charali Chandu (192472332) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

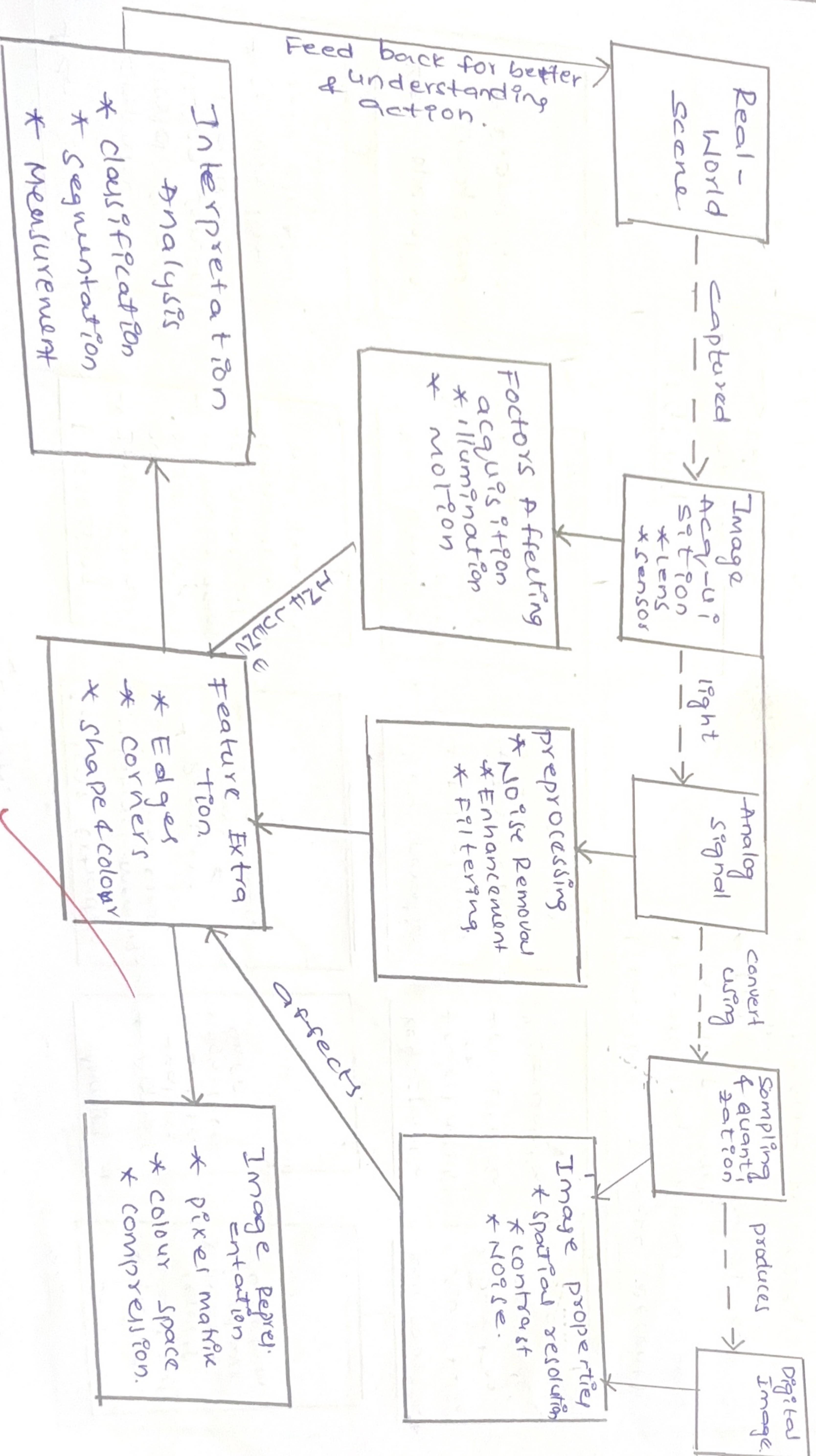
Signature of the Student: C. Chandu

Instructions

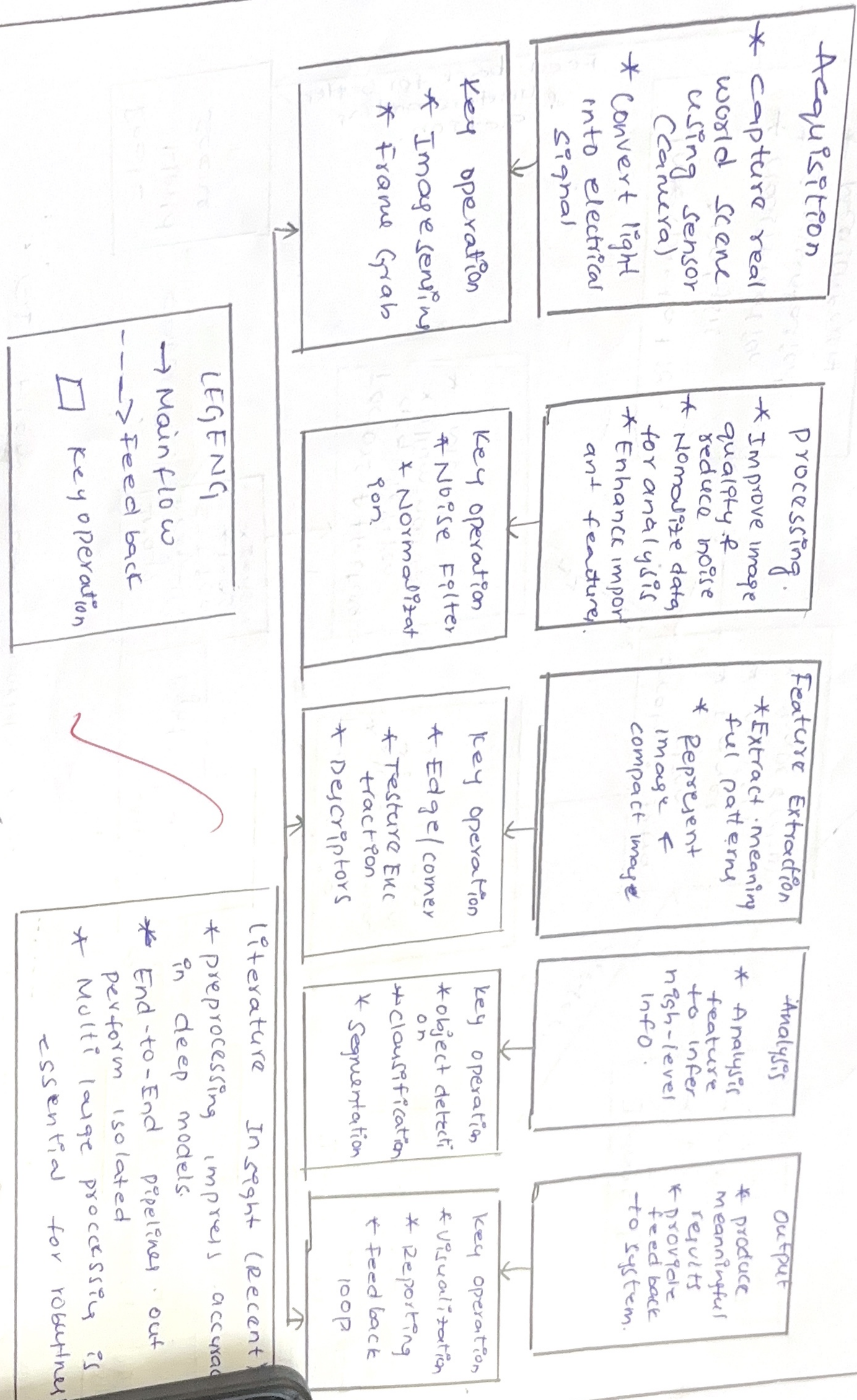
- This test contains 5 questions, each carrying 10 marks. Total: 50 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Overview of Computer Vision, Features of Computer Vision, and Levels of Computer Vision	Computer Vision pipeline, features, and processing levels	1	10
Digital Image Fundamentals	Image representation, resolution, and image quality	2	10
Image Sensing and Acquisition, Sampling and Quantization	Digital image acquisition, sampling, and quantization process	3	10
Image Formation Models	Real-world scene to digital image formation	4	10
Applications of Image Processing and Computer Vision	Computer Vision capabilities and real-world applications	5	10
GRAND TOTAL:			50 Marks

Q1. From Real - World scene Capture to digital Image.



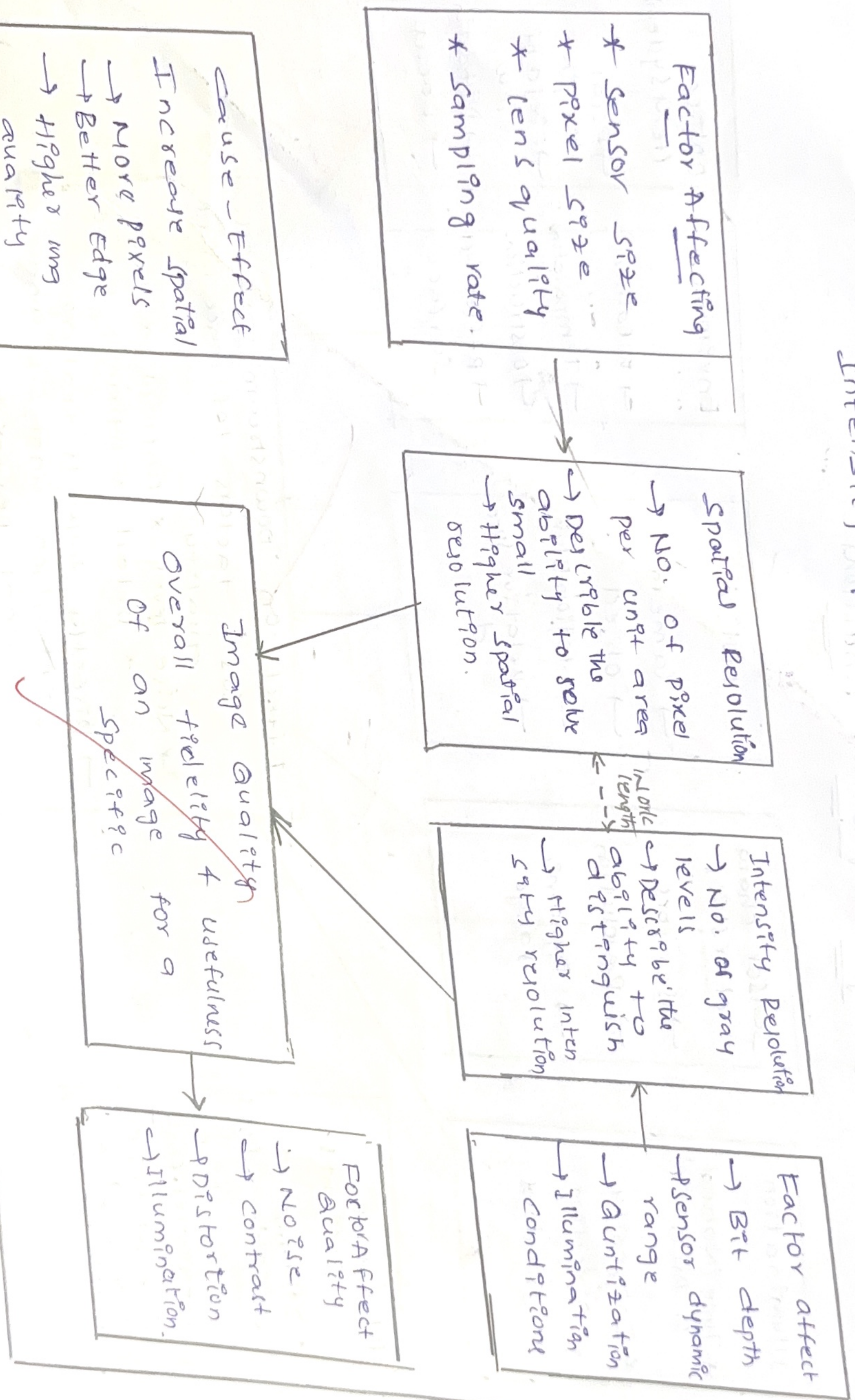
Stages of Visual data Handling in a Computer Vision System.



Literature Insight (Recent)

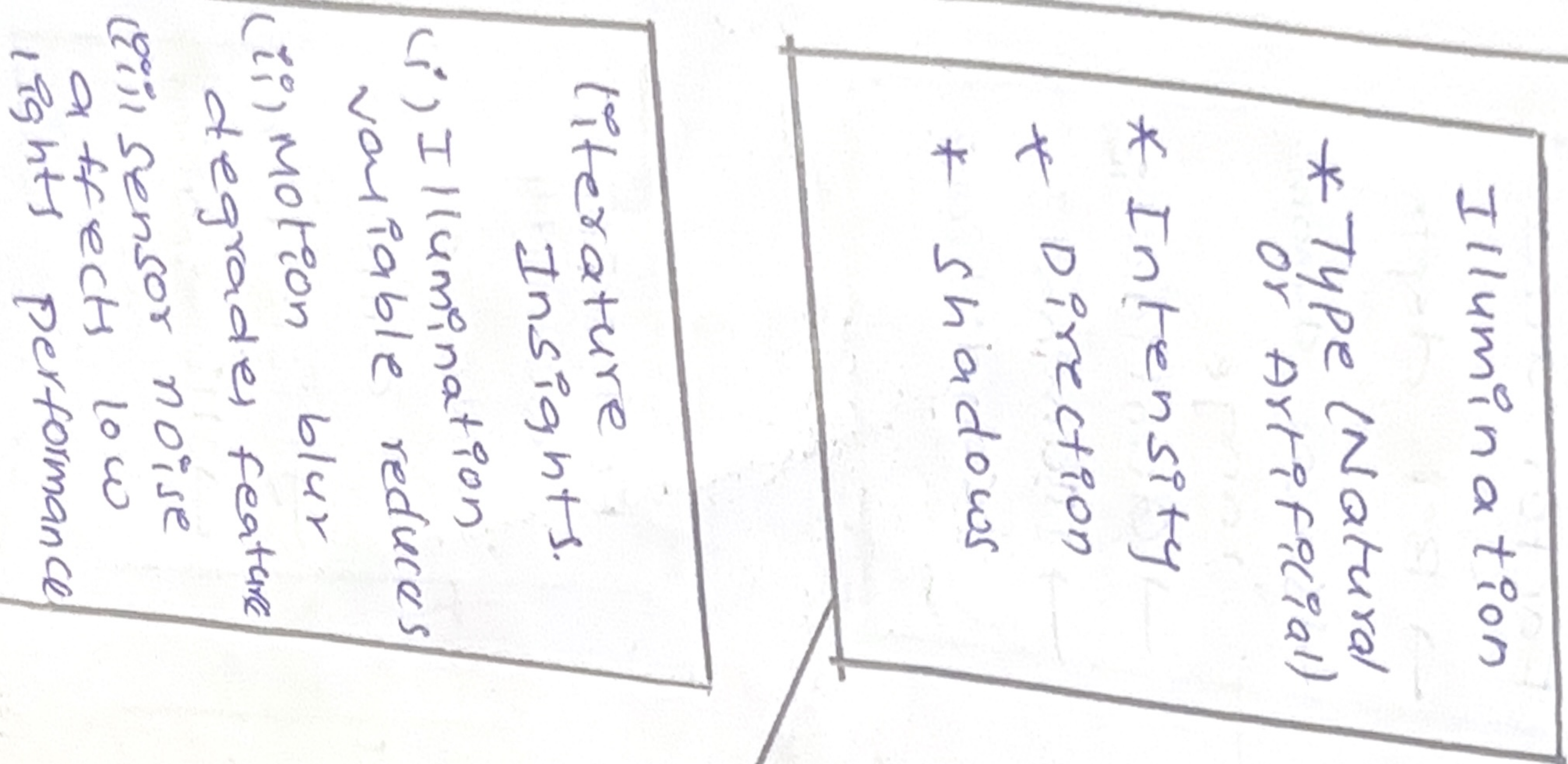
- * Preprocessing improves accuracy in deep models.
- * End-to-End pipelines outperform isolated perform processing is essential for robustness.

Q3: Relationship Among Image Resolution, Spatial Resolution, & Image Quality.



Q4: Factors Influencing Image Acquisition Condition.

Image Acquisition Condition.



Effect on Downstream Process

These factors influence image quality

- * Contrast
- * Sharpness
- * Noise level
- * Detail
- * Dynamic Range

→ Directly impacts feature extraction, analysis and final performance.

Literature Insights.

- (i) Illumination variable reduces
- (ii) Motion blur degrades feature
- (iii) Sensor noise affects low light performance

→ Image
→ Tumor
→ Breast

→ Infrared
→ Spectral
→ Detection

→ Image
→ Tumor
→ Breast

Q5: Computer Vision Capability and Selection of Vision Based Application.

